

Homework 5

Due: Friday 4/18 at 11:59pm

Complete the problem below and submit through Google Classroom. You can submit a text file, Google Doc, etc or complete it on paper and submit pictures or a scan of the paper.

Here are the problems:

1. (10 points) Give the running time in Big-O notation for each of the following. Assume that all arrays, lists, etc have size n . If you want partial credit, include a brief explanation for your answers. Assume that `ArrayList` is implemented the way we saw in class and that `LinkedList` is a doubly-linked list with a tail pointer.
 - (a) The `iterator` method of an `ArrayList<String>`. Recall that this method returns an iterator pointing at the beginning of the list.
 - (b) The `next` method of the iterator returned by the previous part. Recall that this method returns the next value in the list and advances the iterator to the next position.
 - (c) The `contains` method of `LinkedList<String>`, which takes a string and returns whether it appears in the list.
 - (d)

```
int count = 0;
for(int i=0; i < A.length; i++) {
    if(A[i] != 0) {
        count++;
        for(int j=i+1; j < A.length; j++)
            if(A[i] == A[j])
                A[j] = 0;
        A[i] = 0;
    }
}
```
 - (e)

```
for(int i=0; i < A.length; i++) {
    if(A[i] == B[B.length-1-i])
        A[i]++;
    if((i < A.length-1) && (A[i] < A[i+1]))
        A[i]++;
}
```
 - (f)

```
for(int i=0; i < B.length; i++) {
    B[i] = 0;
}
best = 0;
for(int i=0; i < A.length; i++) {
    for(int j=i; j < A.length; j++)
        if(A[i] == A[j])
            B[i]++;
    if(A[i] * B[i] > best)
        best = A[i] * B[i];
}
```
2. (1 point) At the top of your submission, put a statement about collaboration.